

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-02

BOTANY

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Topics To Covered

- 1** 6-Kingdom Classification ✓
- 2** Kingdom Monera (01) ✓

06-KINGDOM CLASSIFICATION

- Introduced by "Carl Woese"
- He divided living organisms into "3 domains" (AKA 3-domain classification).

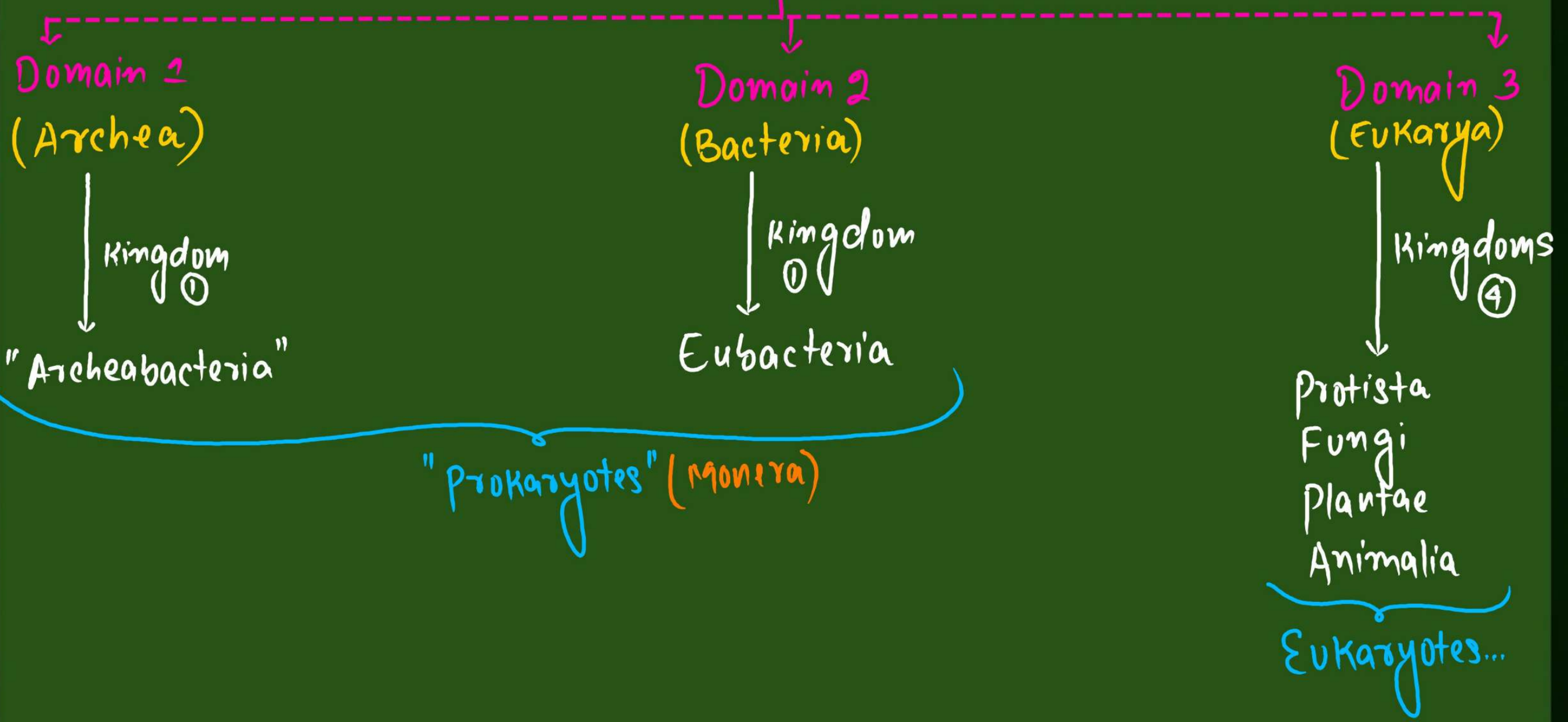
* Domain:-

Domain is above Kingdom Category



All living organisms

3 domains



Note:-

• over time, an attempt has been made to evolve a Classification — which reflects not only morphological, Anatomical, behavioural properties, but also phylogenetic.

NCERT READING



The three-domain system has also been proposed that divides the Kingdom Monera into two domains, leaving the remaining eukaryotic kingdoms in the third domain and thereby a six kingdom classification. You will learn about this system in detail in higher classes.

Over time, an attempt has been made to evolve a classification system which reflects not only the morphological, physiological and reproductive similarities, but is also phylogenetic, i.e., is based on evolutionary relationships.

In this chapter we will study characteristics of Kingdoms Monera, Protista and Fungi of the Whittaker system of classification. The Kingdoms Plantae and Animalia, commonly referred to as plant and animal kingdoms, respectively, will be dealt separately in chapters 3 and 4.

KINGDOM MONERA

≡ Kingdom of prokaryotes.

≡ Bacteria are the sole (i.e., only) members of Kingdom Monera.

Habitat: Cosmopolitan (i.e., everywhere) (Most abundant micro-organisms.)
air, sand, water etc.

Hot Springs, hot deserts, Snow etc.

⊛ A handful of Soil — Contains 100's of bacteria

⊛ Many bacteria live in/on other organisms as parasites.

2.1 KINGDOM MONERA

Bacteria are the sole members of the Kingdom Monera. They are the most abundant micro-organisms. Bacteria occur almost everywhere. Hundreds of bacteria are present in a handful of soil. They also live in extreme habitats such as hot springs, deserts, snow and deep oceans where very few other life forms can survive. Many of them live in or on other organisms as parasites.

SHAPE OF BACTERIA

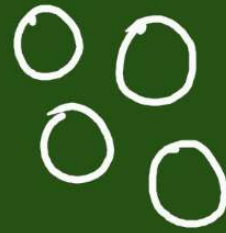
≡ Shape of bacteria — "variable"

≡ On the basis of shape, bacteria — 04 types:

① Coccus (pl = cocci)

→ Spherical

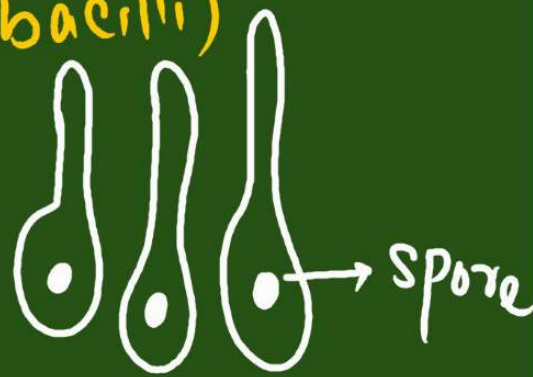
→ Ex. Diplococcus



② Bacillus (pl = bacilli)

→ Rod shaped.

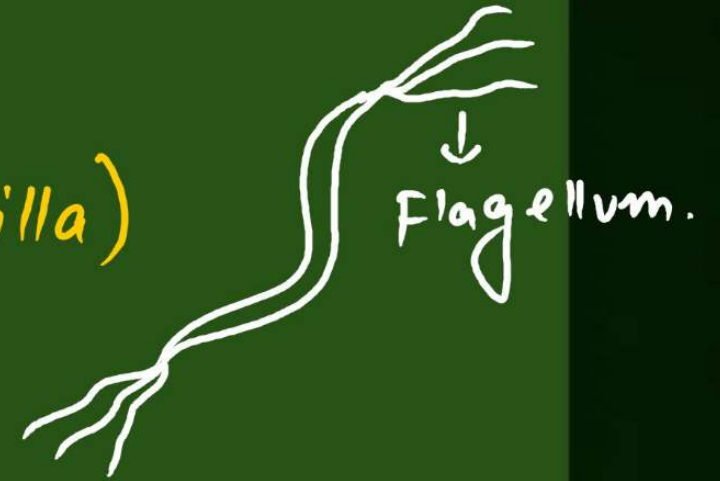
Ex. E. coli



③ Spirillum (pl = spirilla)

→ Spiral shaped

Ex. Spirillum volutans



④ Vibrium (pl = vibrio)

→ Comma shaped

Ex. Vibrio cholerae



NCERT READING

Bacteria are grouped under four categories based on their shape: the spherical Coccus (pl.: cocci), the rod-shaped Bacillus (pl.: bacilli), the comma-shaped Vibrium (pl.: vibrio) and the spiral Spirillum (pl.: spirilla) (Figure 2.1).

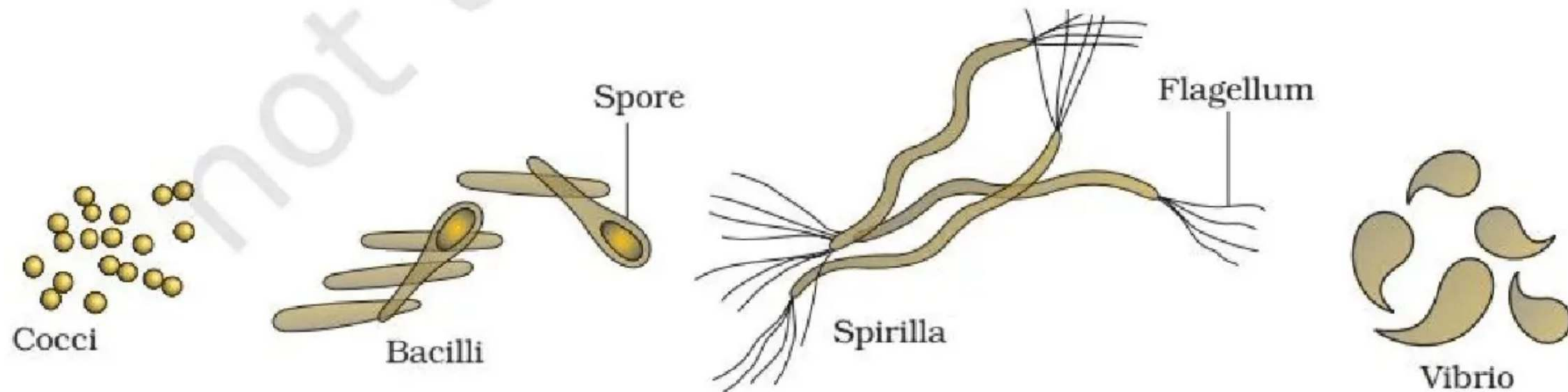
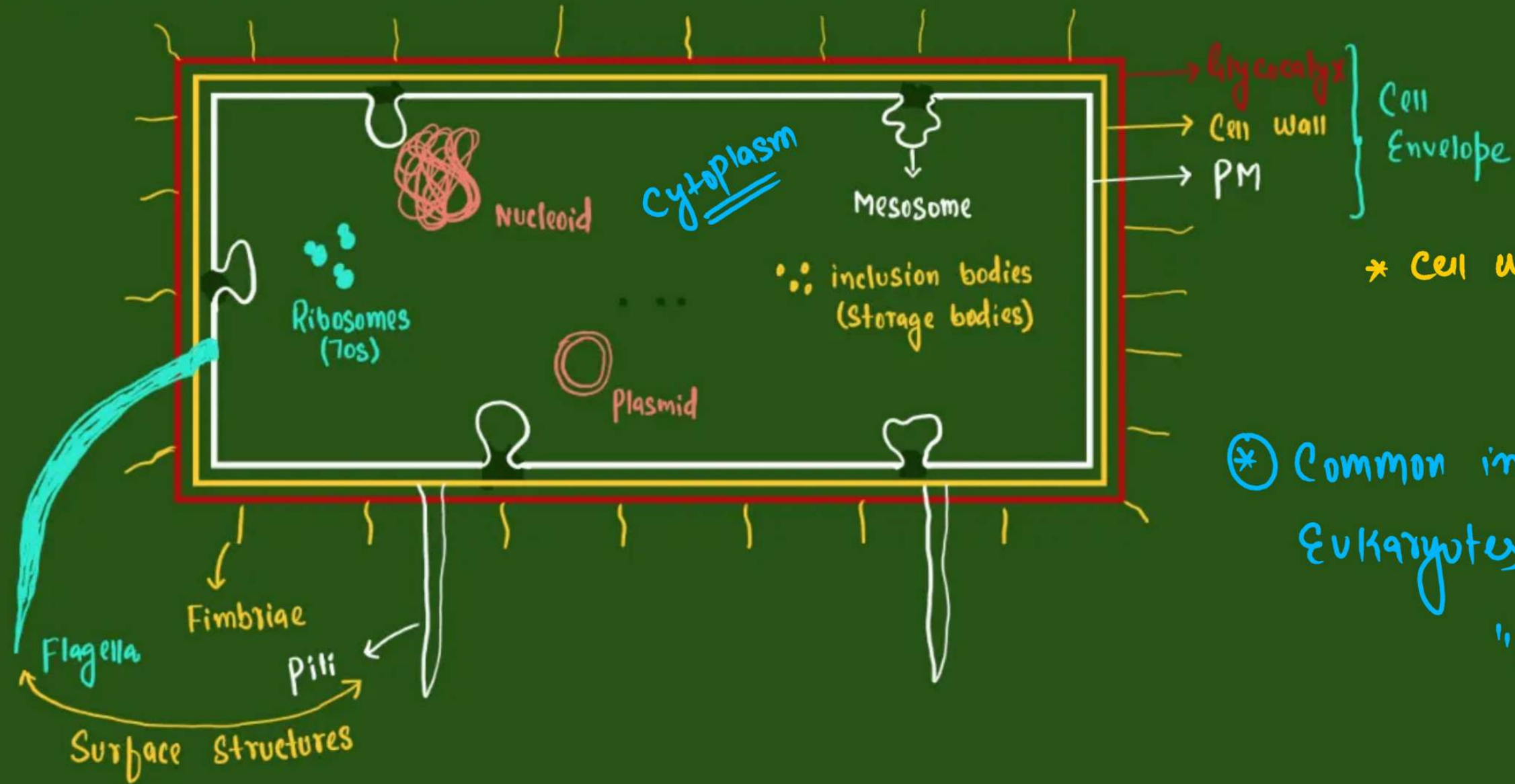


Figure 2.1 Bacteria of different shapes

BACTERIAL CELL STRUCTURE



* Cell wall → made of peptidoglycan...

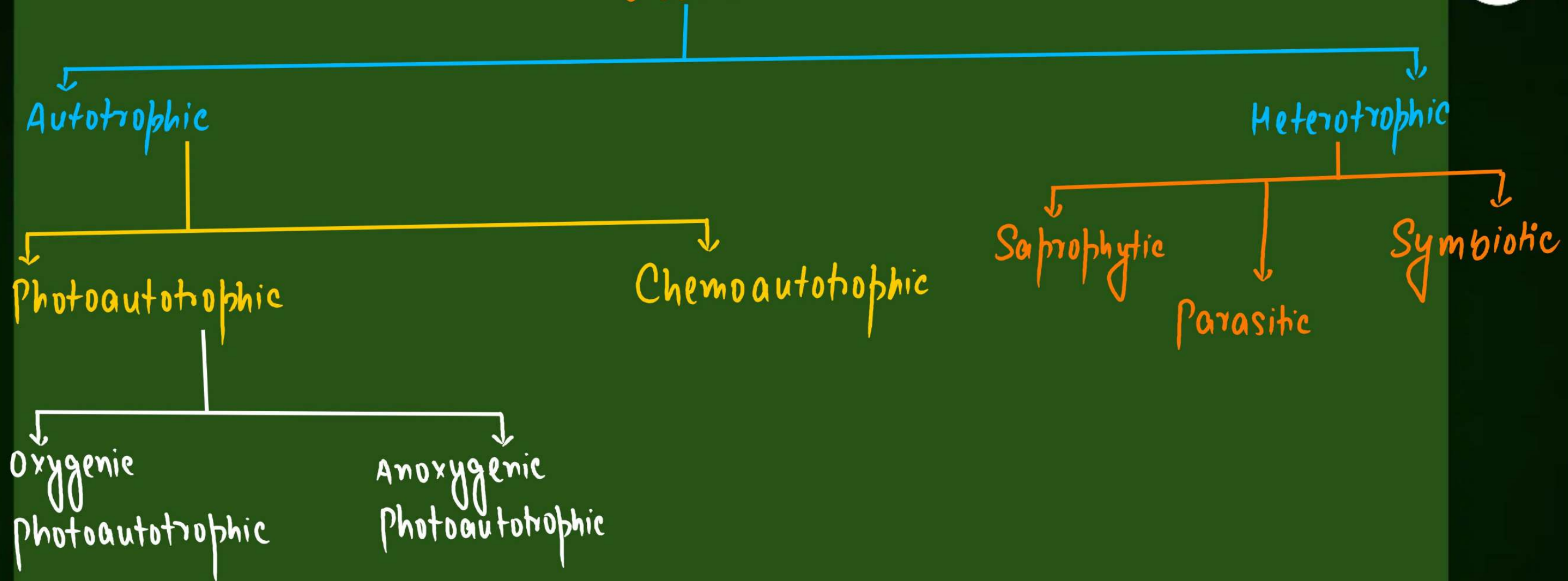
⊛ Common in prokaryotes and eukaryotes → "plasmamembrane"

NUTRITION IN BACTERIA

≡ Bacteria

- Simple in Structure (because they are prokaryotes).
- Complex in behaviour (They show high metabolic diversity).

Nutrition



Note:-

- ①. Heterotrophic bacteria $>$ Autotrophic bacteria.
- ②. Among heterotrophs — Saprophytic bacteria are max.

NCERT READING



Though the bacterial structure is very simple, they are very complex in behaviour. Compared to many other organisms, bacteria as a group show the most extensive metabolic diversity. Some of the bacteria are autotrophic, i.e., they synthesise their own food from inorganic substrates. They may be photosynthetic autotrophic or chemosynthetic autotrophic. The vast majority of bacteria are heterotrophs, i.e., they depend on other organisms or on dead organic matter for food.

QUESTIONS

Which of the following is very simple in structure but very complex in behaviour?

- ☒ 1 Bacteria
- ☐ 2 Archaeobacteria
- ☐ 3 Cyanobacteria
- ☐ 4 Mycoplasma

QUESTIONS

H/W

Sole member of kingdom monera is:

- 1 Archaeobacteria
- 2 Mycoplasma
- 3 Cyanobacteria
- 4 Bacteria

QUESTIONS

H/W

Carl Woese proposed:

- 1 Three-domain system
- 2 Five kingdom classification
- 3 Four kingdom classification
- 4 Two kingdom classification

QUESTIONS

H/w

Match the process in Column I with its term in Column II.

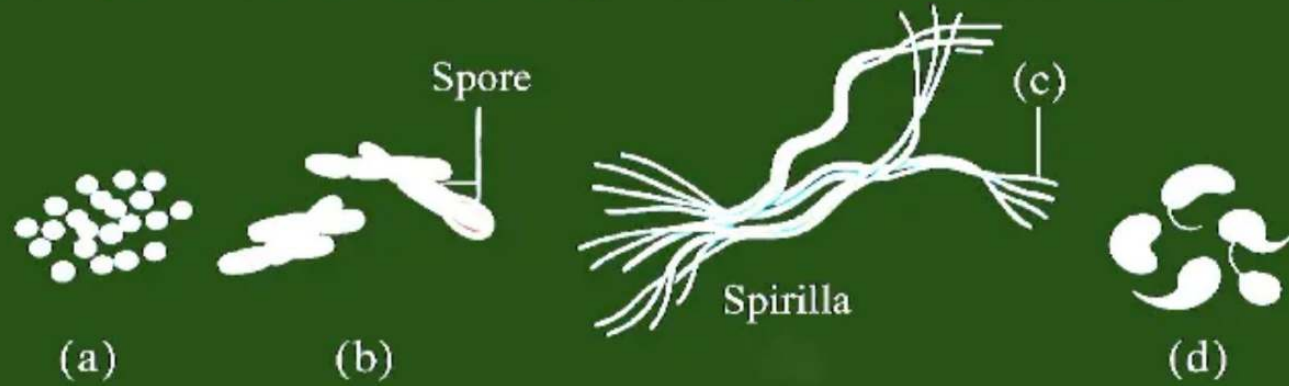
- 1 A–(r); B–(q); C–(p); D–(s)
- 2 A–(s); B–(r); C–(q); D–(p)
- 3 A–(q); B–(p); C–(s); D–(r)
- 4 A–(p); B–(s); C–(r); D–(q)

Column-I (Names)		Column-II (Shape)	
A.	Coccus	p	Rod-shaped
B.	Bacillus	q	Spherical
C.	Vibrio	r	Spiral-shaped
D.	Spirillum	s	Comma-shaped

QUESTIONS

H/W

Label the a, b, c and d in diagram by correct identification of them:



- 1 (a)-Bacilli bacteria, (b)-Vibrio bacteria, (c)-Flagellum in Bacteria, (d)-Cocci bacteria
- 2 (a)-Cocci bacteria, (b)-Bacilli bacteria, (c)-Flagella in Bacteria, (d)-Comma Bacteria
- 3 (a)-Cocci bacteria, (b)-Heterocyst, (c)-Vibrio bacteria, (d)-Bacilli Bacteria
- 4 (a)-Bacilli bacteria, (b)-Pleomorphic bacteria, (c)-Vibrio bacteria, (d)-Cocci Bacteria

Thank
You